

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

For the reaction $A(g) \rightleftharpoons B(g)$ at 495 K,

$$\Delta_r G^\circ = -9.478 \text{ kJ mol}^{-1}$$

If we start the reaction in a closed container at

495 K with 22 millimoles of A, the amount of B in the equilibrium mixture is....millimoles. (Round off to the Nearest Integer). $[R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}; \ln 10 = 2.303]$

Q2: 17 March (Shift 1) - Numerical

The standard enthalpies of formation of Al_2O_3 and CaO are $-1675 \text{ kJ mol}^{-1}$ and -635 kJ mol^{-1} respectively. For the reaction $3CaO + 2Al \rightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy $\Delta_r H^\circ =$ ____ kJ.

(Round off to the Nearest Integer).

Q3: 17 March (Shift 2) - Numerical

Consider the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$.

The temperature at which $K_C = 20.4$ and $K_p = 600.1$, is ____ K. (Round off to the Nearest Integer).

[Assume all gases are ideal and $R = 0.0831 \text{ L bar K}^{-1} \text{ mol}^{-1}$]

Q4: 18 March (Shift 1) - Numerical

For the reaction



the reaction enthalpy $\Delta_r H =$ ____ kJ mol^{-1}

(Round off to the Nearest Integer).

[Given : Bond enthalpies in kJ mol^{-1} : C – C :

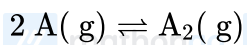
347, C = C : 611; C – H : 414, H – H : 436]

Q5: 18 March (Shift 2) - Numerical

Questions with Answer Keys

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The gas phase reaction



at 400 K has $\Delta G^\circ = +25.2 \text{ kJ mol}^{-1}$.

The equilibrium constant K_C for this reaction is _____ $\times 10^{-2}$. (Round off to the Nearest integer) [Use

$R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln 10 = 2.3$

$\log_{10} 2 = 0.30$, $1 \text{ atm} = 1 \text{ bar}$]

[$\text{antilog}(-0.3) = 0.501$]

Answer Key**Q1 (20)****Q2 (230)****Q3 (354)****Q4 (128)****Q5 (166)**

Q1 (20)

$$\text{Sol. } \Delta G^\circ = -RT \ln K_{eq}$$

$$\text{Given } \Delta G^\circ = -9.478 \text{ kJ/mole}$$

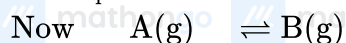
$$T = 495 \text{ K} \quad R = 8.314 \text{ J mol}^{-1}$$

$$\text{So } -9.478 \times 10^3 = -495 \times 8.314 \times \ln K_{eq}$$

$$\ln K_{eq} = 2.303$$

$$= \ln 10$$

$$\text{So } K_{eq} = 10$$



$$t = 0 \quad 22 \quad 0$$

$$t = t \quad 22 - x \quad x$$

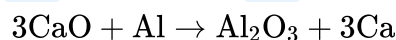
$$K_{eq} = \frac{[B]}{[C]} = \frac{x}{22-x} = 10$$

$$\text{or } x = 20$$

$$\text{So millmoles of B} = 20$$

Q2 (230)

Given reaction:

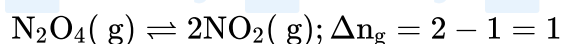


$$\text{Now, } \Delta H^\circ = \sum \Delta_f H^\circ_{\text{Products}} - \sum \Delta_f H^\circ_{\text{Reactants}}$$

$$= [1 \times (-1675) + 3 \times 0] - [3 \times (-635) + 2 \times 0]$$

$$= +230 \text{ kJ mol}^{-1}$$

Q3 (354)



$$\text{Now, } K_p = K_c \cdot (RT)^{\Delta n_g}$$

$$\text{or, } 600.1 = 20.4 \times (0.0831 \times T)^1$$

$$\therefore T = 353.99 \text{ K} = 354 \text{ K}$$

Q4 (128)

$$\Delta_r H = [\epsilon_{C-C} + 2\epsilon_{C-H}] - [\epsilon_{C=C} + \epsilon_{H-H}]$$

$$= [347 + 2 \times 414] - [611 + 436]$$

$$= 128$$

Q5 (166)

Using formula

$$\Delta_r G^0 = -RT \ln K_p$$

$$25200 = -2.3 \times 8.3 \times 400 \log(K_p)$$

$$K_p = 10^{-3.3} = 10^{-3} \times 0.501$$

$$= 5.01 \times 10^{-4} \text{Bar}^{-1}$$

$$= \frac{K_c}{8.3 \times 400}$$

$$K_c = 166 \times 10^{-2} \text{m}^3/\text{mole}$$

Ans = 166